Wind Speed Modelling & Weather Derivatives Pricing

A comprehensive reference — from statistical models to financial applications

Based on Benth & Šaltytė Benth (2009) and Alexandridis & Zapranis (2013)

Reference Sheet — 2026

CONTENTS

1. Introduction & Motivation
2. Statistical Characteristics of Wind Speed
3. Model Survey
    1. Weibull / Rayleigh
    2. AR(p) and ARMA(p,q)
    3. CAR(p) — Continuous-Time OU
    4. Wavelet Networks
    5. ARCH / GARCH
    6. ARFIMA (Long Memory)
    7. Neural Networks / ML
4. Seasonal Components
5. Derivatives Pricing Framework
6. Practical Applications
7. Towards a Better Model
8. Estimation & Diagnostic Checklist
9. References

1 — Introduction & Motivation

Wind speed is the primary driver of wind power production. Its unpredictability exposes wind farm operators to significant revenue risk: periods of low wind reduce production and cash flows, while excessively high speeds force turbines offline for safety. These risks can be transferred using **weather derivatives** — financial contracts whose payoffs depend on wind indices rather than on conventional asset prices.

The market trades both on organised exchanges (Chicago Mercantile Exchange, former US Futures Exchange) and over-the-counter (OTC). Key contracts include:

- **Nordix Wind Speed Index futures:** monthly contracts written on the cumulative deviation of daily wind speed from a 20-year mean, with benchmark 100.
- **Cumulative Average Wind Speed (CAWS) futures:** based on the average wind speed over the measurement period.
- **Options on wind futures:** call and put structures providing flexible hedging.

> WHY MODELLING MATTERS
>
> Building an analytically tractable yet statistically accurate wind speed model is essential for both *risk management* and *fair derivative pricing*. The quality of the model directly determines the quality of the hedge.

2 — Statistical Characteristics of Wind Speed

Distributional Properties

- Raw wind speed is **strictly positive**, right-skewed, and fat-tailed.
- Empirically well described by a **Weibull** distribution; some locations show bimodal behaviour (mixture of two Weibull).
- After a **Box-Cox power transformation** `Y(t) = W(t)^λ` with `λ ≈ 0.1–0.2`, the series becomes approximately normal — a key requirement for closed-form pricing.

$$Y(t) = [W(t)^{\lambda} - 1] / \lambda, \quad \lambda \neq 0 \qquad \text{(Box-Cox)}$$
$$Y(t) = \ln W(t), \quad \lambda = 0$$

Temporal Structure

- **Seasonality in the mean:** clear annual cycle, stronger in winter/early spring, weaker in summer.
- **Seasonality in the variance:** residual variance is also periodic; well-approximated by a truncated Fourier series.
- **Autocorrelation:** significant at lags 1–4 on daily data; PACF cuts off at lag 4 → suggests AR(4) structure.
- **Long memory:** evidence in hourly data; the autocorrelation function decays hyperbolically rather than exponentially.

> PRICING IMPLICATION OF THE TRANSFORMATION
>
> The Box-Cox step is not merely a statistical convenience — it is a **structural requirement**. Closed-form futures and option prices under the risk-neutral measure require Gaussian residuals. Without this transformation, one must resort to full Monte Carlo.

3 — Model Survey

3.1 Weibull / Rayleigh Distribution

Fit a parametric distribution to the *marginal* distribution of wind speed. The Weibull pdf is:

$$f(w; k, c) = (k/c) \cdot (w/c)^{k-1} \cdot \exp[-(w/c)^k], \quad w \geq 0$$

k = shape, c = scale. Rayleigh is the special case k = 2.

PROS	CONS
— Simple estimation (MLE or moments)	— Captures marginal distribution only — no temporal dynamics
— Widely validated across geographies	— Ignores autocorrelation, seasonality, volatility clustering
— Suitable for long-run energy yield assessments	— Cannot be used for derivative pricing (no SDE structure)

3.2 AR(p) and ARMA(p,q)

After deseasonalising and applying the Box-Cox transform, the residual series `W̃(t)` is modelled as:

$$\text{AR}(p): \tilde{W}(t) = \Sigma\, \beta_i\, \tilde{W}(t-i) + \varepsilon(t), \quad \varepsilon(t) \sim N(0, \sigma^2(t))$$

Empirical studies consistently find p = 4 on daily data (Benth 2009, Alexandridis 2013)

$$\text{ARMA}(p,q): \tilde{W}(t) = \Sigma_i\, \beta_i\, \tilde{W}(t-i) + \Sigma_j\, \theta_j\, \varepsilon(t-j) + \varepsilon(t)$$

PROS	CONS
— Parsimonious and easy to estimate	— Linear — cannot capture nonlinear dynamics
— AR(4) yields approximately normal residuals after preprocessing	— MA component adds limited benefit for daily wind
— Directly maps to the continuous-time CAR(p) model	— Constant speed of mean reversion may be unrealistic
— Pricing formulas available in closed form	— Residuals may still exhibit conditional heteroskedasticity

3.3 CAR(p) — Continuous-Time Autoregressive Model

Benth & Šaltytė Benth (2009) embed the AR(p) in a continuous-time stochastic framework. The state vector `X(t) ∈ ℝᵖ` satisfies the vectorial Ornstein-Uhlenbeck SDE:

$$dX(t) = A \cdot X(t)\, dt + e_p\, \sigma(t)\, dB(t)$$

B(t) = standard Brownian motion; σ(t) > 0 = seasonal volatility

Matrix A (p×p):

$$\begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & & \\ 0 & 0 & 0 & \cdots & 1 \\ -\alpha_p & -\alpha_{p-1} & \cdots & & -\alpha_1 \end{pmatrix}$$

The wind speed is recovered via the inverse Box-Cox transform:

$$W(t) = [\Lambda(t) + X_1(t)]^{1/\lambda}, \quad \lambda \neq 0$$
$$W(t) = \exp[\Lambda(t) + X_1(t)], \quad \lambda = 0$$

Λ(t) = deterministic seasonal mean function (truncated Fourier series)

> STATIONARITY CHECK
>
> For p = 4 with New York wind data, the eigenvalues of A are all negative real parts ($\lambda_1 = -1.29$, $\lambda_{2,3} = -0.91 \pm 0.44i$, $\lambda_4 = -0.54$), guaranteeing stationarity.

PROS	CONS
— Rigorous SDE framework; enables risk-neutral pricing via Girsanov	— Requires 1/λ to be a natural number for closed-form pricing
— Futures prices have closed-form Gaussian moment expressions	— Samuelson effect is **violated**: futures volatility may decrease near maturity
— Stationary for p = 4 (all eigenvalues of A have negative real part)	— Constant α_k parameters — time-varying mean reversion not modelled
— Extends naturally to multi-location (spatial) framework	— Gaussian tails may underfit extreme events (storms)

3.4 Wavelet Networks (WN)

Alexandridis & Zapranis (2013) replace the linear AR filter with a nonparametric **wavelet network**, embedded in the same OU stochastic framework:

$$\hat{W}(t) = \beta_0 + \Sigma_k\, \beta_k \cdot \psi\left([\tilde{W}(t-1), \ldots, \tilde{W}(t-n) - b_k] / a_k \right)$$

ψ(·) = wavelet activation; a_k = dilation; b_k = translation parameters

The WN needs only 3 lagged inputs vs. 4 for AR, yet outperforms AR(4) in both in-sample fit and out-of-sample forecasting.

PROS	CONS
— Captures nonlinear dynamics missed by AR	— More parameters — risk of overfitting on short samples
— Consistently lower RMSE / MAE than AR(4)	— Less interpretable; complicates economic intuition
— Universal approximator of continuous functions	— Pricing formulas still require Gaussian residuals
— More parsimonious in lag structure (3 lags suffice)	— Training sensitive to initialisation and wavelet choice

3.5 ARCH / GARCH Models

Tol (1997) proposed modelling conditional heteroskedasticity of wind speed residuals with GARCH:

$$\varepsilon(t) = \sigma(t)\, z(t), \quad z(t) \sim N(0,1)$$
$$\sigma^2(t) = \omega + \Sigma_i\, \alpha_i\, \varepsilon^2(t-i) + \Sigma_j\, \gamma_j\, \sigma^2(t-j)$$

GARCH(1,1) is typically sufficient for wind speed residuals

PROS	CONS
— Captures volatility clustering and time-varying conditional variance	— Complex continuous-time embedding — complicates analytical pricing
— Well-studied estimation (QMLE) and diagnostic tools	— After seasonal detrending, volatility clustering is moderate for wind
— Extensions (EGARCH, GJR-GARCH) allow asymmetric responses	— N(0,1) for z(t) often insufficient; fat-tailed distributions needed

3.6 ARFIMA — Long Memory Models

When wind speed shows hyperbolic autocorrelation decay, ARFIMA(p,d,q) with fractional differencing `d ∈ (0, 0.5)` is appropriate:

$$(1-L)^d\, \Phi(L)\, \tilde{W}(t) = \Theta(L)\, \varepsilon(t)$$

Φ(L) = AR polynomial; Θ(L) = MA polynomial; d = fractional differencing

PROS	CONS
— Captures slow hyperbolic decay of autocorrelation	— Estimation computationally demanding; d estimation is sensitive
— Relevant for hourly data and multi-day forecasting horizons	— Pricing in continuous time with fractional noise is non-trivial
— Outperforms ARMA for longer forecast horizons (Kavasseri 2009)	— Limited to linear dynamics; cannot capture nonlinearities

3.7 Neural Networks / Machine Learning

SVM, RNN/LSTM, and ensemble methods are used primarily for *short-term operational forecasting* (hours-ahead), not for derivative pricing.

PROS	CONS
— Excellent short-term predictive accuracy (1–24h horizon)	— No SDE foundation — cannot directly price derivatives
— Can incorporate exogenous variables (pressure, temperature, humidity)	— Require large datasets; lack interpretability
— LSTM captures long temporal dependencies	— Model uncertainty is difficult to quantify and propagate

4 — Seasonal Components

Seasonal Mean Function

Fitted by a truncated Fourier series (with optional linear trend `γt` for climate drift):

$$\Lambda(t) = a_0 + \gamma t + \Sigma_k \left[a_k \cos(2\pi kt/365) + b_k \sin(2\pi kt/365) \right]$$

Alexandridis & Zapranis (2013) find a slight downward trend in New York wind speeds over the 1988–2008 sample.

Seasonal Variance Function

Computed empirically by grouping AR residuals by calendar day, then fitted as:

$$\sigma^2(t) = c_0 + \Sigma_i\, c_i \sin(2\pi it/365) + \Sigma_j\, d_j \cos(2\pi jt/365)$$

$\sigma^2(t + 365) = \sigma^2(t)$ — periodic with one-year cycle

Pattern (New York data): variance peaks in **winter/early spring**, lowest in **summer** — roughly the inverse of temperature variance seasonality.

5 — Derivatives Pricing Framework

Market Price of Wind Risk

Wind is a *non-traded* underlying — standard no-arbitrage arguments do not uniquely determine the risk-neutral measure. The market price of wind risk `θ(t)` is introduced via Girsanov's theorem:

$$dB(t) = dB^{\theta}(t) + \theta(t)\, dt$$

$B^{\theta}(t)$ = Brownian motion under the risk-neutral measure Q; θ(t) estimated from traded prices

Solution of CAR(p) under Q

Applying Itô's formula to the CAR(p) SDE under the risk-neutral measure:

$$X(s) = \exp(A(s-t))\, X(t) + \int_t^s \exp(A(s-u))\, e_p\, \sigma(u)\, dB^{\theta}(u)$$

$X_1(s)$ is conditionally Gaussian under Q with:

$$\mu^{\theta}(t,s,x) = e_1^{\top} \exp(A(s-t))\, x + \theta \int_t^s e_1^{\top} \exp(A(s-u))\, e_p\, \sigma(u)\, du$$

$$\Sigma^2(t,s) = \int_t^s [e_1^{\top} \exp(A(s-u))\, e_p]^2\, \sigma^2(u)\, du$$

Wind Speed Indices

$$\text{Nordix Index: } N(\tau_1,\tau_2) = 100 + \Sigma_s [W(s) - w_{20}(s)], \quad \tau_1 \leq s \leq \tau_2$$

$w_{20}(s)$ = 20-year historical mean wind speed on day s

$$\text{CAWS Index: } C(\tau_1,\tau_2) = (1/(\tau_2-\tau_1+1)) \cdot \Sigma_s W(s)$$

Futures Pricing Formula

$$F(t; \tau_1,\tau_2) = E^Q[N(\tau_1,\tau_2) \mid F_t] = w_{20}(\tau_1,\tau_2) + \Sigma_s E^Q[W(s) \mid F_t]$$

$E^Q[W(s)|F_t]$ depends on Gaussian moments via inverse Box-Cox

The Samuelson Effect — and Its Violation

In standard OU (CAR(1)) models, futures volatility decays exponentially toward spot volatility as maturity approaches — the **Samuelson effect**. For CAR(p) with p ≥ 2, high-order memory makes the future wind increasingly predictable near maturity, so futures volatility may *decrease* rather than increase (Benth 2009). This has direct consequences for option pricing and delta-hedging strategies.

Options on Wind Futures

$$C(t,T,K) = e^{-r(T-t)} \cdot E^Q[\max(F(T; \tau_1,\tau_2) - K, 0) \mid F_t]$$

- `λ = 0` (log-transform): volatility is deterministic → **Black-76 formula applies**.
- `λ ≠ 0`: stochastic volatility → **Monte Carlo simulation required**.

6 — Practical Applications

1. **Wind power producers:** short wind futures to hedge revenue risk. If the Nordix index falls below 100, the short position compensates for lost production.
2. **Electricity retailers & TSOs:** manage imbalance costs from wind forecast errors. Short-term AR or neural network models feed into operational dispatch.
3. **Insurers and reinsurers:** price weather-linked bonds and cat derivatives using Weibull tails and extreme value theory.
4. **Energy traders:** exploit calendar spreads between wind futures contracts to profit from seasonal mean-reversion anomalies.
5. **Wind farm developers:** lock in minimum revenue via long-dated OTC derivatives during construction and early operation phases.

7 — Towards a Better Model

> CORE IDEA
>
> No single model is optimal across all dimensions. A **modular hybrid architecture** can combine the best features of each class — at the cost of reduced analytical tractability.

1 Richer autoregressive structure: AR → ARMA → ARFIMA

Replace AR(4) with ARMA(4,1) to capture additional MA effects. For hourly data or multi-step forecasting, consider ARFIMA(p,d,q) with d ≈ 0.2–0.35. The MA component may partially substitute for WN nonlinearity while preserving tractability. Use AIC/BIC to assess whether the gain is significant — on daily data the improvement is often marginal (Torres et al. 2005).

2 Time-varying mean reversion speed κ(t)

In both Benth (2009) and Alexandridis (2013), the mean reversion speed is constant. Allowing κ(t) to be a periodic function captures the empirical finding that wind autocorrelation varies seasonally:

$$dX_1(t) = \kappa(t)[\Lambda(t) - X_1(t)]\, dt + \sigma(t)\, dB(t)$$

Pricing formulas remain tractable if κ(t) is a step function or a truncated Fourier series. This was explored via time-varying AR coefficients by Ailliot et al. (2006).

3 Two-component GARCH volatility

The seasonal Fourier variance $\sigma^2(t)$ captures only the unconditional seasonal pattern. Residual volatility clustering is captured by a multiplicative GARCH layer:

$$\sigma^2_{\text{total}}(t) = \tilde{\sigma}^2(t) \cdot h(t)$$
$$h(t) = \omega + \alpha\, \varepsilon^2(t-1) + \beta\, h(t-1)$$

$\tilde{\sigma}^2(t)$ is the seasonal baseline; h(t) is the GARCH(1,1) multiplier. This structure is analogous to Tol (1997) but keeps the seasonal envelope explicit. For pricing, h(t) can be approximated as piecewise-constant.

4 Stochastic volatility — Heston-type adaptation

Replace deterministic σ(t) with a stochastic variance process:

$$dX_1(t) = \kappa[\Lambda(t) - X_1(t)]\, dt + \sqrt{V(t)}\, dB_1(t)$$
$$dV(t) = \eta[\bar{v}(t) - V(t)]\, dt + \xi \sqrt{V(t)}\, dB_2(t)$$

$\rho = \text{Corr}(dB_1, dB_2)$; $\bar{v}(t)$ = seasonal mean-reversion level for variance

Futures prices remain semi-analytically tractable via characteristic function methods. Options require numerical methods (FFT, Monte Carlo).

5 Lévy-driven noise for fat tails

Replace Brownian motion with a Lévy process L(t) (e.g., Normal Inverse Gaussian or Variance Gamma) to account for extreme wind events:

$$dX(t) = A \cdot X(t)\, dt + e_p\, \sigma(t)\, dL(t)$$

The CAR(p) framework with Lévy noise is studied in Brockwell & Marquardt (2005). Pricing formulas involve the moment generating function of L(t) rather than Gaussian moments, and remain tractable for NIG and VG processes.

6 WN + GARCH + Lévy — Full nonparametric hybrid

Combine all extensions: wavelet network autocorrelation, seasonal GARCH variance, and Lévy innovations:

$$\tilde{W}(t) = \text{WN}(\tilde{W}(t-1), \ldots, \tilde{W}(t-n)) + \sigma(t)\sqrt{h(t)} \cdot z(t)$$

This captures nonlinear dynamics (WN), seasonal heteroskedasticity, volatility clustering (GARCH), and fat tails (Lévy/t-innovations). Derivative pricing requires full Monte Carlo under Q. This is the most accurate but least tractable option — best suited to risk management and VaR, not real-time hedging.

Model Comparison Table

MODEL	SEASONAL	NONLINEAR	VOL. CLUSTER	FAT TAILS	TRACTABLE	PRICING
Weibull	—	—	—	✓	✓	—
AR(p)	✓	—	—	—	✓	✓
ARMA(p,q)	✓	—	—	—	✓	✓
ARFIMA	✓	—	—	—	(✓)	—
CAR(p) / OU	✓	—	—	—	✓	✓
CAR + GARCH	✓	—	✓	—	(✓)	(✓)
WN (Alexandridis)	✓	✓	—	—	✓	✓
Heston-wind	✓	—	✓	—	(✓)	(✓)
CAR + Lévy	✓	—	—	✓	✓	✓
WN + GARCH + Lévy	✓	✓	✓	✓	—	MC only

(✓) = partially tractable; MC = Monte Carlo only. Highlighted row is the full hybrid model.

> TRACTABILITY-FLEXIBILITY TRADE-OFF
>
> The richer the model, the more it relies on simulation for pricing. For practical hedging, closed-form or semi-analytical formulas are preferred. For risk management (VaR, CVaR), a full Monte Carlo with a rich model is more appropriate.

8 — Estimation & Diagnostic Checklist

1. Apply Box-Cox transform; estimate λ by maximum likelihood.
2. Fit and subtract seasonal mean Λ(t) — Fourier regression with optional linear trend γt.
3. Check ACF and PACF of deseasonalised series; determine p (and q for ARMA).
4. Fit AR/ARMA or WN to residuals; verify whiteness via Ljung-Box Q-test.
5. Inspect ACF of **squared** residuals for seasonal heteroskedasticity; fit $\sigma^2(t)$ via Fourier series.
6. After dividing by σ(t), test for remaining GARCH effects (ARCH-LM test). If significant, fit GARCH(1,1).
7. Test residual normality — Jarque-Bera, Kolmogorov-Smirnov, QQ-plot. If strongly rejected, consider Lévy innovations (NIG, Student-t).
8. Conduct out-of-sample forecasting evaluation (RMSE, MAE, MAPE) across multiple horizons (1-day, 5-day, 2-month).
9. If data from multiple stations are available, consider a spatial-temporal extension (Šaltytė Benth et al. 2007).

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